

The Dirichlet Pólya Problem for Spherical Shells in General Dimension: Exact Dimension Lifting and the Shifted-Tail Bottleneck

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Abstract

For a spherical shell in dimension $d \geq 3$, the separated radial problem has Bessel orders $\ell + (d-2)/2$ and spherical-harmonic multiplicities $\kappa_{d,\ell}$. We give an exact branching decomposition of the strict shell-action proxy into positive integer- and half-integer shifted planar tails. We also prove a signed primitive comparison and retain its full positive remainder. This yields the exact identity

$$W_d - P_d^< = \mathcal{B}_d(A) + \sum_{m \geq 0} \binom{m+d-3}{d-3} D_A \left(m + \frac{d-2}{2} \right).$$

Consequently, one dimension-free positive shifted-tail inequality implies the sharp Dirichlet Pólya inequality for every spherical shell and every $d \geq 3$. The dimension lift is proved here; the inner shifted-tail inequality remains the central open lemma. We prove the outer and terminal tail regions, close the first interface-cell milestone by exact rational wall estimates, and prove all shifted tails above an explicit fixed-ratio high-energy threshold. A wall-safe first-shelf reduction compresses every remaining many-cell tail to one scalar coupled to its terminal one-interface reserve. In the thin sector it also improves the uniform threshold to $K \geq 125/[8(1-\rho)^2]$. A scale-free shelf-curvature bound and a level-by-level inverse-action estimate give an explicit exact-angle lower bound for the terminal reserve. No assertion in this manuscript treats the residual terminal-shelf scalar as proved.

1 Statement and status

For $0 < \rho < 1$ let

$$A_{\rho,1}^{(d)} = \{x \in \mathbb{R}^d : \rho < |x| < 1\}, \quad d \geq 3,$$

and use the strict counting convention

$$N_D(A_{\rho,1}^{(d)}, K^2) := \#\{j : \lambda_j^D(A_{\rho,1}^{(d)}) < K^2\}.$$

Write

$$\omega_d = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}, \quad L_d = \frac{\omega_d}{(2\pi)^d}.$$

The desired estimate is

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq L_d |A_{\rho,1}^{(d)}| K^d = \frac{1 - \rho^d}{2^d \Gamma(d/2 + 1)^2} K^d. \tag{1}$$

For $a > 0$, define the zero-extended ball action

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z \leq a, \\ 0, & z > a, \end{cases} \quad (2)$$

put $\mu = \rho K$, and set

$$A(z) = A_{\rho, K}(z) := G_K(z) - G_\mu(z). \quad (3)$$

For $x > 0$, let

$$[x]_< := \#\{n \in \mathbb{N} : n < x\} = \max\{0, [x] - 1\}.$$

In particular, $[n]_< = n - 1$ at a positive integral wall.

The only unproved statement on the primary path is the following.

Conjecture 1.1 (Positive shifted shell tails). *For $0 < \rho < 1$, $K > 0$, and*

$$r \in \tfrac{1}{2}\mathbb{N},$$

one has

$$T_A(r) := [A(r) + \tfrac{1}{4}]_< + 2 \sum_{j \geq 1} [A(r + j) + \tfrac{1}{4}]_< \leq 2 \int_r^K A(z) \, dz. \quad (4)$$

The zero extension makes the sum finite and makes the assertion trivial when $r \geq K$.

Sections 2–5 prove that Conjecture 1.1 implies (1) in every dimension $d \geq 3$. Section 6 records the proved tail regions, the completed first interface-cell theorem, and a fixed-ratio high-energy theorem.

2 General-dimensional spectral proxy

Put

$$a_d = \frac{d-2}{2}, \quad \nu_{\ell, d} = \ell + a_d.$$

The degree- ℓ spherical harmonics on $\mathbb{S}^{d-1}$ have multiplicity

$$\kappa_{d, \ell} = \binom{\ell + d - 1}{d - 1} - \binom{\ell + d - 3}{d - 1} = \frac{(2\ell + d - 2)(\ell + d - 3)!}{\ell!(d - 2)!}. \quad (5)$$

Here and below a binomial coefficient with lower index larger than its nonnegative upper index is zero.

Under the unitary radial substitution

$$u(r) = r^{(d-1)/2} R(r),$$

the degree- ℓ block becomes

$$L_{\ell, d} = -\frac{d^2}{dr^2} + \frac{\nu_{\ell, d}^2 - \frac{1}{4}}{r^2} \quad \text{on } (\rho, 1), \quad u(\rho) = u(1) = 0. \quad (6)$$

Its positive eigenfrequencies are the zeros of

$$D_{\nu, \rho}(k) = J_\nu(\rho k) Y_\nu(k) - Y_\nu(\rho k) J_\nu(k), \quad \nu = \nu_{\ell, d}.$$

With the standard continuous Bessel phase, let

$$\gamma_{\rho,K}(\nu) = \frac{\theta_\nu(K) - \theta_\nu(\rho K)}{\pi}.$$

The determinant factorization and phase monotonicity give the exact strict count

$$N_D(A_{\rho,1}^{(d)}, K^2) = \sum_{\ell \geq 0} \kappa_{d,\ell} [\gamma_{\rho,K}(\nu_{\ell,d})]_{<}. \quad (7)$$

The real-order phase-difference enclosure of [2, 3] gives, for $0 \leq \nu \leq K$,

$$\gamma_{\rho,K}(\nu) < A_{\rho,K}(\nu) + \frac{1}{4}. \quad (8)$$

For $\nu \geq K$, both the exact count and the action proxy vanish. Since $x < y$ implies $[x]_{<} \leq [y]_{<}$, including when y is integral, (7)–(8) imply the wall-safe proxy bound

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq P_d^<(\rho, K), \quad P_d^< := \sum_{\ell \geq 0} \kappa_{d,\ell} q_\ell, \quad (9)$$

where

$$q_\ell = [A(\ell + a_d) + \frac{1}{4}]_{<}. \quad (10)$$

The argument above is dimension-independent after the substitutions $\nu = \ell + a_d$ and (5). In particular, the published phase enclosure is already a real-order result: no interpolation is introduced when d changes parity.

3 Exact harmonic branching

Let $p = d - 2$, so $a_d = p/2$, and define

$$r_m = m + \frac{p}{2}, \quad c_{d,m} = \binom{m+p-1}{p-1} = \binom{m+d-3}{d-3}. \quad (11)$$

Proposition 3.1 (Exact shifted-tail decomposition). *For every finitely supported sequence $(q_\ell)_{\ell \geq 0}$,*

$$\sum_{\ell \geq 0} \kappa_{d,\ell} q_\ell = \sum_{m \geq 0} c_{d,m} \left(q_m + 2 \sum_{j \geq 1} q_{m+j} \right). \quad (12)$$

Consequently, for (10),

$$\boxed{P_d^< = \sum_{m \geq 0} c_{d,m} T_A(r_m)}. \quad (13)$$

This identity is exact at every strict action wall.

Proof. The generating functions are

$$\sum_{\ell \geq 0} \kappa_{d,\ell} s^\ell = \frac{1+s}{(1-s)^{d-1}}, \quad \sum_{m \geq 0} c_{d,m} s^m = \frac{1}{(1-s)^{d-2}},$$

and

$$1 + 2 \sum_{j \geq 1} s^j = \frac{1+s}{1-s}.$$

Coefficient comparison gives

$$\kappa_{d,\ell} = c_{d,\ell} + 2 \sum_{m=0}^{\ell-1} c_{d,m}.$$

Substitution and finite reindexing prove (12). For (10), one has $q_{m+j} = [A(r_m + j) + 1/4]_<$, which proves (13). No floor has been approximated or moved. $\square$

The shifts are

$$r_m \in \begin{cases} \mathbb{N}_0 + \frac{1}{2}, & d \text{ odd}, \\ \mathbb{N}, & d \text{ even}. \end{cases}$$

Thus Conjecture 1.1 is asked only on the two grids that actually occur.

4 The signed Weyl primitive

Define the right-continuous branching staircase

$$S_d(z) = \sum_{\substack{m \geq 0 \\ r_m \leq z}} c_{d,m} \tag{14}$$

and its discrepancy primitive

$$\Delta_d(z) = \int_0^z \left(S_d(u) - \frac{u^p}{p!} \right) du. \tag{15}$$

Proposition 4.1 (Signed branching primitive). *For every $d \geq 3$ and $z \geq 0$,*

$$\boxed{\Delta_d(z) \leq 0.} \tag{16}$$

For $d = 3$, equality at positive z occurs precisely at the positive integers. For $d \geq 4$, the inequality is strict for $z > 0$.

Proof. For $0 \leq z < r_0 = p/2$, $S_d(z) = 0$, so

$$\Delta_d(z) = -\frac{z^{p+1}}{(p+1)!} \leq 0.$$

Let $M \geq 0$ and $r_M = M + p/2$. The hockey-stick identity gives

$$\int_0^{r_M} S_d(u) du = \sum_{m=0}^{M-1} (r_M - r_m) c_{d,m} = \binom{M+p}{p+1}.$$

Therefore

$$(p+1)! \Delta_d(r_M) = \prod_{j=0}^p (M+j) - \left(M + \frac{p}{2} \right)^{p+1} \leq 0, \tag{17}$$

because the arithmetic mean of $M, M+1, \dots, M+p$ is $M + p/2$.

On the half-open cell $[r_M, r_{M+1})$,

$$S_d(z) = \binom{M+p}{p}, \quad \Delta'_d(z) = \binom{M+p}{p} - \frac{z^p}{p!}.$$

If the critical point lies in this cell, it is

$$z_M^* = \left(\prod_{j=1}^p (M+j) \right)^{1/p}.$$

Direct substitution into the cell formula gives

$$(p+1)! \Delta_d(z_M^*) = p(z_M^*)^p \left(z_M^* - M - \frac{p+1}{2} \right) \leq 0, \quad (18)$$

since z_M^* is the geometric mean of $M+1, \dots, M+p$ and their arithmetic mean is $M + (p+1)/2$. If the critical point is outside the cell, the cell maximum is at an endpoint, already covered by (17). This proves (16).

For $p = 1$, the cell maximum is $z = M + 1$ and has value zero. For $p \geq 2$, the consecutive factors in both AM–GM applications are unequal, which gives the stated strictness. $\square$

5 Exact defect identity and conditional closure

Define

$$D_A(r) = 2 \int_r^K A(z) dz - T_A(r) \quad (19)$$

and the continuous payment

$$W_d = \frac{2}{p!} \int_0^K z^p A(z) dz. \quad (20)$$

Lemma 5.1 (Exact Weyl normalization). *For $p = d - 2$,*

$$\int_0^a z^p G_a(z) dz = \frac{\Gamma((p+1)/2)}{4\sqrt{\pi}(p+2)\Gamma((p+4)/2)} a^{p+2}. \quad (21)$$

Consequently,

$$W_d = \frac{1 - \rho^d}{2^d \Gamma(d/2 + 1)^2} K^d = L_d |A_{\rho,1}^{(d)}| K^d. \quad (22)$$

Proof. Scale $z = ax$ in (2), integrate the term containing $\arccos x$ by parts, and evaluate the resulting beta integrals. This gives (21). Multiplying the difference of (21) at $a = K$ and $a = \rho K$ by $2/p!$, the duplication formula for Γ reduces the coefficient to

$$\frac{1}{2^{p+2} \Gamma((p+4)/2)^2} = \frac{1}{2^d \Gamma(d/2 + 1)^2}.$$

The last equality in (22) follows from $|A_{\rho,1}^{(d)}| = \omega_d(1 - \rho^d)$. $\square$

Proposition 5.2 (Branching-defect identity). *Let*

$$\mathcal{B}_d(A) = 2 \int_0^K (-A'(z))(-\Delta_d(z)) dz. \quad (23)$$

Then $\mathcal{B}_d(A) \geq 0$ and

$$\boxed{W_d - P_d^< = \mathcal{B}_d(A) + \sum_{m \geq 0} c_{d,m} D_A(r_m).} \quad (24)$$

Proof. The shell action is absolutely continuous, decreasing, and vanishes at K . Fubini gives

$$\sum_{m \geq 0} c_{d,m} 2 \int_{r_m}^K A(z) dz = 2 \int_0^K S_d(z) A(z) dz.$$

Since $\Delta'_d = S_d - z^p/p!$ almost everywhere and both endpoint terms vanish, integration by parts gives

$$2 \int_0^K S_d A = W_d + 2 \int_0^K A \Delta'_d = W_d - 2 \int_0^K A' \Delta_d = W_d - \mathcal{B}_d(A).$$

Both A' and Δ_d are nonpositive, so $\mathcal{B}_d(A) \geq 0$. Subtract (13) and use (19). $\square$

Theorem 5.3 (Conditional all-dimensional shell theorem). *If Conjecture 1.1 holds, then (1) holds for every $d \geq 3$, $0 < \rho < 1$, and $K \geq 0$. It then also holds for every shell $A_{r,R}^{(d)} = \{x \in \mathbb{R}^d : r < |x| < R\}$ and every spectral parameter.*

Proof. The case $K = 0$ is trivial, so assume $K > 0$. Every r_m in (11) lies in $\frac{1}{2}\mathbb{N}$, so the conjecture gives $D_A(r_m) \geq 0$. Equations (9), (24), and (22) yield

$$N_D(A_{\rho,1}^{(d)}, K^2) \leq P_d^< \leq W_d = L_d |A_{\rho,1}^{(d)}| K^d.$$

For a shell with outer radius R , take $\rho = r/R$ and $K = R\sqrt{\Lambda}$ and use Dirichlet dilation. The non-strict counting form follows by applying the strict estimate at $\Lambda + \varepsilon$ and letting $\varepsilon \downarrow 0$. $\square$

6 Proved tail regions and the first interface cell

The action derivative is

$$-A'(z) = \frac{1}{\pi} \begin{cases} \arccos \frac{z}{K} - \arccos \frac{z}{\mu}, & 0 < z < \mu, \\ \arccos \frac{z}{K}, & \mu \leq z < K. \end{cases} \quad (25)$$

Thus A is decreasing and concave on $(0, \mu)$, decreasing and convex on (μ, K) , and A, A' match at μ .

Proposition 6.1 (Outer and terminal tails). *Conjecture 1.1 holds in either of the following cases:*

1. $r \geq \mu$;
2. $G_K(r) \leq 3/4$.

Proof. If $r \geq \mu$, then $A = G_K$ on $[r, K]$; the claim is trivial when $r \geq K$. Otherwise set $B = \lceil K - r \rceil$ and zero-extend

$$g(s) = G_K(r + s) \quad (0 \leq s \leq K - r)$$

to the integer interval $[0, B]$. Since $G_K(K) = G'_K(K) = 0$, this extension is nonnegative, decreasing, convex, $1/2$ -Lipschitz, and vanishes at B . The convex shifted-floor theorem of [1] therefore bounds its ordinary-floor tail by $2 \int_0^B g$. At every sample,

$$[g(j) + 1/4]_< \leq \lfloor g(j) + 1/4 \rfloor.$$

The strict tail is consequently bounded by $2 \int_0^B g = 2 \int_r^K G_K$.

If $G_K(r) \leq 3/4$, monotonicity and $A \leq G_K$ imply $A(r + j) + 1/4 \leq 1$ for every $j \geq 0$. Every strict bracket is therefore zero. Notice that equality at 1 is harmless precisely because the bracket is strict. $\square$

We next sharpen the prescribed first milestone. For the ball action set

$$T_K(r) = [G_K(r) + \tfrac{1}{4}]_< + 2 \sum_{j \geq 1} [G_K(r + j) + \tfrac{1}{4}]_<, \quad D_K(r) = 2 \int_r^K G_K(z) dz - T_K(r). \quad (26)$$

The wall-safe translation in the preceding proof gives $D_K(r) \geq 0$.

Lemma 6.2 (A low-action $1/3$ -Lipschitz reserve). *Let g be strictly decreasing and convex on $[0, b]$, with $g(b) = 0$ and $\text{Lip}(g) \leq 1/3$. If*

$$\frac{3}{4} < g(0) < 1,$$

then

$$2 \int_0^b g(z) dz - \left([g(0) + \tfrac{1}{4}]_< + 2 \sum_{j \geq 1} [g(j) + \tfrac{1}{4}]_< \right) > \frac{11}{16}. \quad (27)$$

The function is understood to be zero-extended after b .

Proof. Let $h = g(0)$ and let $x \in (0, b)$ be defined by $g(x) = 3/4$. Since $g(j) < 1$, a sample contributes exactly one if and only if $j < x$. Therefore the bracketed count is

$$2\lceil x \rceil - 1.$$

The Lipschitz bound and $g(b) = 0$ give

$$g(z) \geq \max\{h - z/3, 0\}, \quad 2 \int_0^b g(z) dz \geq 3h^2 > \frac{27}{16}.$$

If $x \leq 1$, the count is one and (27) follows.

Suppose $x > 1$ and put $a = (h - 3/4)/x$. Convexity implies that, for $z \geq x$,

$$g(z) \geq \max\left\{\frac{3}{4} - a(z - x), 0\right\}.$$

Hence

$$\int_0^b g(z) dz \geq \frac{3}{4}x + \frac{(3/4)^2}{2a} = x \left(\frac{3}{4} + \frac{9}{32(h - 3/4)} \right) > \frac{15}{8}x.$$

Since $2\lceil x \rceil - 1 < 2x + 1$, the defect is larger than

$$\frac{15}{4}x - (2x + 1) = \frac{7}{4}x - 1 > \frac{3}{4} > \frac{11}{16}.$$

$\square$

Proposition 6.3 (Exact one-interface reduction). *Assume*

$$0 < \mu - r < 1, \quad r \in \frac{1}{2}\mathbb{N}.$$

Define

$$n_0 = [G_K(r) + \frac{1}{4}]_{<} - [G_K(r) - G_\mu(r) + \frac{1}{4}]_{<} \in \mathbb{N}_0 \quad (28)$$

and

$$\delta_\mu(r) = \int_r^\mu G_\mu(z) \, dz.$$

Then

$$\boxed{D_A(r) = D_K(r) + n_0 - 2\delta_\mu(r).} \quad (29)$$

Moreover,

$$2\delta_\mu(r) < \frac{4(\mu - r)^{5/2}}{15\sqrt{\mu}} < \frac{4\sqrt{2}}{15} < 1. \quad (30)$$

Consequently, the shifted-tail inequality holds if any of the following is true:

1. $n_0 \geq 1$;
2. $G_K(\max\{r, K/2\}) \geq 1$;
3. $r \geq K/2$.

The only remaining one-interface configurations satisfy

$$\boxed{r < \frac{K}{2}, \quad K < \frac{1}{\omega_0}, \quad n_0 = 0, \quad \omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}.} \quad (31)$$

In particular,

$$r \in \left\{ \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2} \right\}.$$

Proof. Since $r + j > \mu$ for every $j \geq 1$, all samples after the first agree with the ball samples. Thus $T_A(r) = T_K(r) - n_0$. Subtracting $2 \int_r^\mu G_\mu$ from the ball integral proves (29).

The turning estimate

$$G_\mu(z) < \frac{(\mu - z)^{3/2}}{3\sqrt{\mu}} \quad (0 < z < \mu)$$

integrates to (30), using $\mu > r \geq 1/2$ and $\mu - r < 1$. If $n_0 \geq 1$, then $D_K(r) \geq 0$ and (29) is positive.

For $r < K$, use the same integer endpoint $B = \lceil K - r \rceil$ and zero-extended function $g(s) = G_K(r + s)$. Put

$$s_* = \max\{0, K/2 - r\}.$$

Then g is $1/3$ -Lipschitz on $[s_*, B]$ and $g(s_*) = G_K(\max\{r, K/2\})$. The refined ordinary-floor theorem of [3, Theorem 3.4], followed by $[\cdot]_{<} \leq \lfloor \cdot \rfloor$, gives

$$D_K(r) \geq \frac{1}{2} \lfloor G_K(\max\{r, K/2\}) \rfloor. \quad (32)$$

For $r \geq K$ the same inequality is trivial. Hence the second condition and (30) prove the claim.

If $r \geq K/2$ and $G_K(r) < 1$, the translated ball action is globally $1/3$ -Lipschitz. Its tail is either zero or Lemma 6.2 applies. In the latter case $D_K(r) > 11/16 > 4\sqrt{2}/15$, so (29) is again positive. This proves the third condition.

If none of the three conditions holds, then $r < K/2$ and $G_K(K/2) = \omega_0 K < 1$, giving (31). The elementary rational bounds

$$\frac{265}{153} < \sqrt{3} < \frac{26}{15}, \quad \frac{333}{106} < \pi < \frac{355}{113}$$

(the square-root bounds follow by squaring) give $1/10 < \omega_0 < 1/9$. Consequently

$$\frac{9}{2} < \frac{1}{2\omega_0} < 5.$$

Since $r < K/2 < 1/(2\omega_0)$ and $r \in \frac{1}{2}\mathbb{N}$, only the nine displayed shifts remain. $\square$

Remark 6.4 (The residual first-interface family). Proposition 6.3 replaces the only unresolved part of the first-interface problem by a bounded nine-shift family: $2r < K < 1/\omega_0$, $r < \mu < \min\{r+1, K\}$, and $n_0 = 0$. The inequality to be checked there is exactly

$$D_K(r) \geq 2\delta_\mu(r).$$

Proposition 6.5 and Lemma 6.6 below close this family.

Proposition 6.5 (Finite ball-wall reduction). *In the residual family of Proposition 6.3, put*

$$B = G_K(r) + \frac{1}{4}, \quad q = [B]_<, \quad e = B - q \in (0, 1],$$

and set

$$E_r(K) = D_K(r) - e.$$

Then

$$2\delta_\mu(r) < e. \tag{33}$$

Consequently, the residual one-interface inequality follows from $E_r(K) \geq 0$.

On every open K -interval containing no ball-sample wall, E_r is strictly increasing. At a wall of the first sample $G_K(r) + 1/4$, the two unit drops in $D_K(r)$ and e cancel, so E_r is continuous. At a wall of a later sample $G_K(r+j) + 1/4$, $j \geq 1$, E_r drops by two for each crossed level. It is therefore enough to check the right-hand limits at the finitely many later-sample walls, together with the first positive-count wall and any lower endpoint $K = 2r$ at which the count is already positive.

More explicitly, all walls that can occur have the form

$$G_K(r+j) = n - \frac{1}{4}, \quad j \geq 0, \quad n \in \{1, 2, 3\}, \quad r+j < 10. \tag{34}$$

Proof. Let $h = G_\mu(r)$. If $q = 0$, strict increase of $G_a(r)$ in the radius a gives $h < G_K(r) < B = e$. If $q \geq 1$, write $B = q + e$ and use the strict bracket convention; then $[B-h]_< = [B]_<$ holds exactly when $h < e$. Thus, in both cases,

$$n_0 = 0 \iff [B-h]_< = [B]_< \iff h < e.$$

When $q \geq 1$, equality $h = e$ lands on the integral wall q and causes a strict-count drop. The function G_μ is strictly convex on $[r, \mu]$, with endpoint values h and 0. It lies strictly below its endpoint chord, and therefore

$$2\delta_\mu(r) < h(\mu - r) < e,$$

which proves (33). If $T_K(r) = 0$, then $T_A(r) = 0$ and the desired inequality is immediate. Otherwise it is enough, by (29), to prove $D_K(r) \geq e$.

Away from sample walls the integer-valued tail and q are constant. Writing $x = r/K$, differentiation of the ball action and its tail integral gives

$$\pi E'_r(K) = K \left(\arccos x - x\sqrt{1-x^2} \right) - \sqrt{1-x^2}, \quad (35)$$

and

$$\pi E''_r(K) = \arccos x + x\sqrt{1-x^2} - \frac{x^2}{K\sqrt{1-x^2}}. \quad (36)$$

Here $x < 1/2$ and $K > 1$. Hence

$$\pi E''_r(K) > \frac{\pi}{3} - \frac{1}{2\sqrt{3}} > 0.$$

For $r \geq 1$, evaluation at $K = 2r$ gives

$$\pi E'_r(2r) = \frac{2\pi r}{3} - \frac{\sqrt{3}}{2}(r+1) > 0.$$

For $r = 1/2$, positivity of the tail count implies $G_K(r) > 3/4$. Since G_K is $1/2$ -Lipschitz and vanishes at K , this forces $K > 2$. Formula (35) is already positive at $(r, K) = (1/2, 2)$, since

$$\pi E'_{1/2}(2) = 2 \arccos \frac{1}{4} - \frac{3\sqrt{15}}{8} > \frac{2\pi}{3} - \frac{3}{2} > 0.$$

Equation (36) then keeps it positive. Thus E_r is strictly increasing between walls.

The stated jump rules follow directly from the weights $1, 2, 2, \dots$ in $T_K(r)$. At a first-sample wall, e equals 1 and its right-hand limit is 0, exactly matching the unit right-hand drop of D_K . Simultaneous later-sample crossings are combined at the same wall abscissa. Finally, $K < 1/\omega_0 < 10$ and $G_K(z) < K/\pi < 10/3$. Thus only $n = 1, 2, 3$ and samples $r + j < 10$ can occur, and

$$\partial_K G_K(z) = \frac{1}{\pi} \sqrt{1 - \frac{z^2}{K^2}} > 0 \quad (K > z)$$

shows that every pair (j, n) produces at most one wall. This proves the finite reduction. $\square$

Lemma 6.6 (Exact residual wall certificate). *In the residual family of Proposition 6.5, if $T_K(r) > 0$, then*

$$E_r(K) > 0. \quad (37)$$

More precisely, $E_r(K) > 3/20$ for $r \leq 4$, while for $r = 9/2$ one has

$$E_{9/2}(K) > \frac{1253}{555}.$$

Proof. Put

$$H_r(K) = 2 \int_r^K G_K(z) dz - G_K(r) - \frac{1}{4}, \quad M_r(K) = \sum_{j \geq 1} [G_K(r+j) + \frac{1}{4}]_<.$$

The first-sample bracket cancels from the definition of E_r , including at its strict walls, and hence

$$E_r(K) = H_r(K) - 2M_r(K). \quad (38)$$

For $x = r/K$, direct integration gives

$$H_r(K) = \frac{A}{2} - \frac{A \arcsin x + B\sqrt{1-x^2}}{\pi} - \frac{1}{4}, \quad A = \frac{K^2}{2} + r^2 + r, \quad B = K \left(\frac{3r}{2} + 1 \right). \quad (39)$$

We record rational envelopes sufficient to check every wall. For $0 \leq x \leq 13/20$, let

$$a_m = \frac{\binom{2m}{m}}{4^m(2m+1)}, \quad b_m = \frac{\binom{2m}{m}}{4^m(2m-1)},$$

and set

$$L(x) = \sum_{m=0}^6 a_m x^{2m+1}, \quad U(x) = L(x) + \frac{x^{15}}{1-x^2},$$

$$V(x) = 1 - \sum_{m=1}^6 b_m x^{2m}, \quad W(x) = V(x) - \frac{x^{14}}{1-x^2}.$$

The power series, whose omitted positive coefficients are smaller than one, give

$$L \leq \arcsin x \leq U, \quad W \leq \sqrt{1-x^2} \leq V.$$

Together with $333/106 < \pi < 355/113$, they imply

$$\begin{aligned} \underline{G}(K, z) &= K \left[\frac{113}{355} (W + xL) - \frac{x}{2} \right] \leq G_K(z) \\ &\leq K \left[\frac{106}{333} (V + xU) - \frac{x}{2} \right] = \overline{G}(K, z), \end{aligned} \quad (40)$$

where now $x = z/K$, and

$$\frac{A}{2} - \frac{106}{333} (AU + BV) - \frac{1}{4} \leq H_r(K) \leq \frac{A}{2} - \frac{113}{355} (AL + BW) - \frac{1}{4}. \quad (41)$$

Thus every comparison below is a rational-number comparison.

Let $k_n(z)$ denote the unique solution of $G_K(z) = n-1/4$. Substitution in (40) gives the following pairwise disjoint brackets:

wall	bracket	wall	bracket
$k_1(3/2)$	$(22/5, 9/2)$	$k_1(2)$	$(5, 26/5)$
$k_1(5/2)$	$(57/10, 29/5)$	$k_1(3)$	$(63/10, 32/5)$
$k_1(7/2)$	$(69/10, 7)$	$k_1(4)$	$(15/2, 38/5)$
$k_2(3/2)$	$(77/10, 39/5)$	$k_1(9/2)$	$(81/10, 41/5)$
$k_2(2)$	$(83/10, 17/2)$	$k_1(5)$	$(87/10, 44/5)$
$k_2(5/2)$	$(9, 227/25)$		

(40a)

These are all later-sample walls. Indeed, the same rational bounds give

$$\frac{227}{25} < \frac{1}{\omega_0} < \frac{459}{50}$$

and, at $K = 459/50$,

$$\overline{G}(K, 11/2) < \frac{18}{25} < \frac{3}{4}, \quad \overline{G}(K, 3) < \frac{8}{5} < \frac{7}{4}, \quad \overline{G}(K, 3/2) < \frac{9}{4} < \frac{11}{4}.$$

Monotonicity in K and z excludes, respectively, every omitted level-3/4, level-7/4, and level-11/4 or higher wall.

At a right-hand wall limit, the value of M_r is the event number in the following exact order:

r	$(z, n; M_r)$
$\frac{1}{2}$	$(\frac{3}{2}, 1; 1), (\frac{5}{2}, 1; 2), (\frac{7}{2}, 1; 3), (\frac{3}{2}, 2; 4), (\frac{9}{2}, 1; 5), (\frac{5}{2}, 2; 6)$
1	$(2, 1; 1), (3, 1; 2), (4, 1; 3), (2, 2; 4), (5, 1; 5)$
$\frac{3}{2}$	$(\frac{5}{2}, 1; 1), (\frac{7}{2}, 1; 2), (\frac{9}{2}, 1; 3), (\frac{5}{2}, 2; 4)$
2	$(3, 1; 1), (4, 1; 2), (5, 1; 3)$
$\frac{5}{2}$	$(\frac{7}{2}, 1; 1), (\frac{9}{2}, 1; 2)$
3	$(4, 1; 1), (5, 1; 2)$
$\frac{7}{2}$	$(\frac{9}{2}, 1; 1)$
4	$(5, 1; 1)$
$\frac{9}{2}$	$\emptyset.$

Equations (39)–(41) then give the strict rational bounds

r	first-wall lower	first-wall upper	activation or $K = 2r$	other-wall lower
$\frac{1}{2}$	$> 3/20$	$< 8/25$	$> 11/25$	$> 13/20$
1	$> 17/100$	$< 1/2$	$> 14/25$	$> 7/10$
$\frac{3}{2}$	$> 9/25$	$< 53/100$	$> 69/100$	$> 37/50$
2	$> 2/5$	$< 29/50$	$> 37/50$	$> 39/50$
$\frac{5}{2}$	$> 23/50$	$< 63/100$	$> 22/25$	$> 81/100$
3	$> 13/25$	$< 69/100$	$> 47/50$	$> 17/20$
$\frac{7}{2}$	$> 29/50$	$< 19/25$	$> 109/100$	—
4	$> 33/50$	$< 83/100$	$> 163/100$	— .

For each $r \leq 4$, the two first-wall columns concern the right-hand value at $G_K(r+1) = 3/4$. Its displayed upper bound is smaller than both competing lower bounds, so this is the controlling wall. Formula (35) shows that H_r , and hence E_r between its later-sample walls, is increasing throughout the positive-count residual region. For $r \leq 3$ the tail is zero before its first-sample activation; for $r = 7/2, 4$ the displayed $K = 2r$ value owns the lower endpoint. Finally, for $r = 9/2$ there is no later wall and

$$E_{9/2}(9) = \frac{43}{2} - \frac{279\sqrt{3}}{8\pi} > \frac{1253}{555}.$$

This proves (37). A dependency-free replay of all 628 rational comparisons is supplied in `scripts/general_d_finite_wall_fraction_verify.py`; it is an audit of, not a numerical premise for, the displayed estimates. $\square$

Theorem 6.7 (One-interface shifted tails). *Conjecture 1.1 holds whenever*

$$0 < \mu - r < 1.$$

It also holds at the endpoint $r = \mu$ by Proposition 6.1.

Proof. The three regions in Proposition 6.3 are already proved there. In its residual family, if $T_K(r) = 0$, then $T_A(r) = 0$ and the claim is immediate. Otherwise Lemma 6.6 and Proposition 6.5 give

$$D_K(r) = E_r(K) + e > e > 2\delta_\mu(r).$$

The exact identity (29) now gives $D_A(r) > 0$. $\square$

Proposition 6.8 (Fixed-ratio high-energy tails). *For $0 < \rho < 1$, put*

$$c_\rho = \frac{\arccos \rho}{\pi}, \quad d_\rho = 1 - 2c_\rho = \frac{2 \arcsin \rho}{\pi},$$

$$\eta_\rho = G_1 \left(\max \left\{ \rho, \frac{1}{2} \right\} \right), \quad a_\rho = \frac{2\pi\rho}{1-\rho}, \quad C_* = \frac{1}{2} + \frac{8\sqrt{2}}{15},$$

and

$$K_0(\rho) = \frac{a_\rho + 2\eta_\rho C_* + \sqrt{a_\rho^2 + 4a_\rho \eta_\rho C_* + \eta_\rho^2}}{2\eta_\rho^2}. \quad (42)$$

If $K \geq K_0(\rho)$, then Conjecture 1.1 holds for every $r \in \frac{1}{2}\mathbb{N}$.

Proof. Only $r < \mu$ requires proof. Set

$$n = \lfloor \mu - r \rfloor, \quad q = r + n, \quad B = \lceil K - r \rceil.$$

Thus $q \leq \mu < q + 1$. For integers $a < b$, write

$$\mathbf{T}(f; a, b) = \frac{1}{2} \lfloor f(a) \rfloor + \sum_{j=a+1}^{b-1} \lfloor f(j) \rfloor + \frac{1}{2} \lfloor f(b) \rfloor.$$

The strict brackets in T_A are bounded above by ordinary floors. Additivity at the split point and $A \leq G_K$ therefore give, when $n \geq 1$,

$$\frac{T_A(r)}{2} \leq \mathbf{T} \left(A(r + \cdot) + \frac{1}{4}; 0, n \right) + \mathbf{T} \left(G_K(r + \cdot) + \frac{1}{4}; n, B \right). \quad (43)$$

For $n = 0$, the first term is absent and the same inequality follows by direct pointwise majorization.

Assume first $n \geq 1$ and put

$$h(t) = A(r + t) + \frac{1}{4}, \quad m_j = \lfloor h(j) \rfloor.$$

If the nonincreasing sequence $(m_j)_{j=0}^n$ leaves its first shelf, let p be its last index on that shelf; otherwise put $p = n$. On $[0, n]$, h is decreasing, concave, and c_ρ -Lipschitz. The quantitative concave trapezoidal-floor theorem [3, Theorem 1.4], with the basic concavity bound in the no-drop case, yields

$$\mathbf{T}(h; 0, n) \leq \int_0^n A(r + t) dt + \frac{n}{4} - \frac{1 - c_\rho}{2} (n - p). \quad (44)$$

For $n = 0$, take $p = 0$ and read this as an absent identity.

Let $\sigma = -A'$. On $0 < z < \mu$,

$$\sigma'(z) = \frac{1}{\pi \sqrt{\mu^2 - z^2}} - \frac{1}{\pi \sqrt{K^2 - z^2}} \geq \kappa := \frac{1 - \rho}{\pi \rho K}.$$

Indeed, after subtracting the value at $z = 0$, the remaining bracket is decreasing in its radius parameter. Since $\sigma(0) = 0$,

$$A(x) - A(y) = \int_x^y \sigma(z) dz \geq \frac{\kappa}{2} (y^2 - x^2) \geq \frac{\kappa}{2} (y - x)^2.$$

The samples at r and $r + p$ lie on the same floor shelf, so their action drop is smaller than one. Keeping the r -term in the preceding estimate and using $r \geq 1/2$ gives

$$p < \sqrt{r^2 + a_\rho K} - r \leq \sqrt{a_\rho K + \frac{1}{4}} - \frac{1}{2}. \quad (45)$$

For the convex suffix, set

$$s = \max \left\{ q, \frac{K}{2} \right\}, \quad t_* = s - r.$$

Then $t_* \in [n, B]$, the translated ball action is $1/2$ -Lipschitz on $[n, B]$ and $1/3$ -Lipschitz on $[t_*, B]$, and

$$G_K(s) \geq K\eta_\rho.$$

The quantitative convex theorem [3, Theorem 3.4] gives

$$\mathbf{T} \left(G_K(r + \cdot) + \frac{1}{4}; n, B \right) \leq \int_n^B G_K(r + t) dt - \frac{1}{4} \lfloor K\eta_\rho \rfloor. \quad (46)$$

The only mismatch between the two integrals on the right of (43) and the shell integral is

$$\delta = \int_q^\mu G_\mu(z) dz.$$

The turning bound from [3, Lemma 6.2] and $0 \leq \mu - q < 1$, $\mu > 1/2$, give

$$0 \leq \delta \leq \frac{2(\mu - q)^{5/2}}{15\sqrt{\mu}} < \frac{2\sqrt{2}}{15}. \quad (47)$$

Combining (43), (44), and (46) gives

$$\frac{T_A(r)}{2} \leq \int_r^K A(z) dz + \delta - \frac{M}{4}, \quad M = \lfloor K\eta_\rho \rfloor + d_\rho(n - p) - p. \quad (48)$$

By (45),

$$M > K\eta_\rho - \sqrt{a_\rho K + \frac{1}{4}} - \frac{1}{2}.$$

The definition (42) is exactly the positive-root condition

$$K\eta_\rho - \sqrt{a_\rho K + \frac{1}{4}} \geq C_*.$$

Thus $M > 8\sqrt{2}/15 > 4\delta$, and (48) proves the result. All inequalities use ordinary floors on the majorizing side, so strict action walls require no limiting argument. $\square$

The fixed-ratio threshold is deliberately uniform in the shift, but it is not sharp as $\rho \uparrow 1$. Retaining the location of the first shelf improves the thin-shell scale by two powers of $1 - \rho$.

Lemma 6.9 (Optimized thin first-shelf loss). *Let*

$$\rho = 1 - \varepsilon, \quad 0 < \varepsilon \leq \frac{1}{100}, \quad K\varepsilon^2 \geq \frac{125}{8}.$$

Use n, p from the proof of Proposition 6.8, put $m = n - p$, and set

$$d = d_\rho, \quad \mathcal{R} = p - dm.$$

If $\mathcal{R} > 0$, then

$$\mathcal{R} < \frac{361}{80\sqrt{\varepsilon}}. \quad (49)$$

Proof. This is a scale-free consequence of the exact local slope and the same-floor condition. Put

$$y = \sqrt{\varepsilon}, \quad \varkappa = K\varepsilon^2, \quad P = py, \quad \mathfrak{r} = \mathcal{R}y, \quad S = (p + m)y.$$

Elementary trigonometric bounds give

$$d > \frac{9}{10}, \quad S = P + \frac{P - \mathfrak{r}}{d} < \frac{19}{9}P. \quad (50)$$

Write

$$U = \mu - r, \quad t = \frac{r}{\mu}, \quad w = \frac{U}{\mu} = 1 - t,$$

and define

$$\mathcal{Q} = 1 + \frac{y^2}{\varkappa} + \frac{yS}{\varkappa}, \quad \hat{q} = \frac{y^2}{\rho}(\mathcal{Q} - 1).$$

The unconditional first-shelf bound $p^2 < 2\pi\rho K/\varepsilon$ and (50) give

$$w < \frac{1 + p + m}{\rho K} = \frac{y^4 + y^3 S}{\varkappa \rho} = \hat{q} < \frac{y^4}{\varkappa \rho} + \frac{19}{9}y\sqrt{\frac{2\pi}{\varkappa \rho}}.$$

Here the first term is at most $8/1237500 < 1/1000$, while the square of the factor following $19/9$ is at most $352/86625 < 1/225$. Hence

$$w < \hat{q} < \frac{1}{1000} + \frac{19}{135} < \frac{1}{7}, \quad t > 1 - \hat{q} > \frac{6}{7}. \quad (51)$$

Using

$$-A'(z) = \frac{1}{\pi} \left(\arccos \frac{z}{K} - \arccos \frac{z}{\mu} \right)$$

in the same-floor inequality $A(r) - A(r + p) < 1$ gives, after scaling,

$$P^2 < \mathcal{H}(P, \mathfrak{r}) := \frac{2\pi^2 \mathcal{Q}}{(1 - \hat{q})^2}. \quad (52)$$

On the preceding domain, $\mathcal{H}$ decreases in $\mathfrak{r}$ and

$$\partial_P \mathcal{H} < \frac{673816}{1366875} < \frac{1}{2}.$$

Let $B_* = 361/80$. If $\mathfrak{r} \geq B_*$, then $P \geq \mathfrak{r}$ and $P^2 - \mathcal{H}(P, B_*)$ is increasing for $P \geq B_*$. Indeed, every intermediate $P' \in [B_*, P]$ retains the unconditional shelf ceiling and satisfies $S(P', B_*) < 19P'/9$, so (51) and the displayed derivative bound hold throughout the comparison interval. Its minimum over $y \leq 1/10$ and $\varkappa \geq 125/8$ is at $P = B_*$, $y = 1/10$, and $\varkappa = 125/8$. There

$$\mathcal{Q} = \frac{12869}{12500}, \quad \hat{q} = \frac{41}{137500},$$

and, using $\pi < 22/7$,

$$B_*^2 - 2 \left(\frac{22}{7} \right)^2 \frac{12869/12500}{(1 - 41/137500)^2} = \frac{72581986185449}{5925464687161600} > 0.$$

This contradicts (52) and proves (49). All displayed comparisons are rational once the elementary bound for π is inserted; the dependency-free script `computations/round9_verify_thin_plateau_constants.py` replays the finite arithmetic. $\square$

Proposition 6.10 (Thin high-energy shifted tails). *Let $\rho = 1 - \varepsilon$ with $0 < \varepsilon \leq 1/100$. If*

$$K \geq \frac{125}{8\varepsilon^2}, \quad (53)$$

then Conjecture 1.1 holds for every $r \in \frac{1}{2}\mathbb{N}$.

Proof. The outer tails are covered by Proposition 6.1. For an inner tail, retain the notation of Proposition 6.8. Before its final estimate, (48) gives

$$\frac{T_A(r)}{2} \leq \int_r^K A(z) dz + \delta - \frac{M}{4}, \quad M = \lfloor KG_1(\rho) \rfloor - \mathcal{R},$$

with $4\delta < 8\sqrt{2}/15$. If $\mathcal{R} \leq 0$, the convex gain already gives $M > 1 > 4\delta$. If $\mathcal{R} > 0$, Lemma 6.9,

$$G_1(1 - \varepsilon) \geq \frac{2\sqrt{2}}{3\pi}\varepsilon^{3/2}, \quad \sqrt{2} > \frac{140}{99}, \quad \sqrt{2} < \frac{99}{70}, \quad \pi < \frac{1571}{500},$$

and $\lfloor x \rfloor > x - 1$ give

$$M > \frac{2829397}{3732696} > \frac{132}{175} > \frac{8\sqrt{2}}{15} > 4\delta.$$

Thus (48) proves the claim. The proof uses only a positive shift and a unit-spaced sampling grid; it is unchanged on the integer and half-integer grids, and every threshold equality is owned by the strict-floor majorant. $\square$

Corollary 6.11 (A uniform small-hole sector). *If*

$$0 < \rho < \omega_0, \quad K(\omega_0 - \rho) \geq \frac{1}{2} + \frac{8\sqrt{2}}{15},$$

then Conjecture 1.1 holds for every $r \in \frac{1}{2}\mathbb{N}$.

Proof. For $r < \mu$, use the split (43), basic concavity on the head, and (46) with $s = K/2$. This gives

$$\frac{T_A(r)}{2} \leq \int_r^K A(z) dz + \delta - \frac{\lfloor \omega_0 K \rfloor - n}{4}.$$

Since $r \geq 1/2$,

$$\lfloor \omega_0 K \rfloor - n > K(\omega_0 - \rho) - \frac{1}{2} \geq \frac{8\sqrt{2}}{15} > 4\delta.$$

The outer tails were proved in Proposition 6.1. $\square$

7 Cell recurrence and the corrected global target

For $x_j = r + j$, set

$$Q_j = [A(x_j) + \frac{1}{4}]_<, \quad D_j = 2 \int_{x_j}^K A(z) dz - \left(Q_j + 2 \sum_{k>j} Q_k \right).$$

Then

$$D_j - D_{j+1} = 2 \int_{x_j}^{x_{j+1}} A(z) dz - Q_j - Q_{j+1}. \quad (54)$$

If

$$e_j = A(x_j) + \frac{1}{4} - Q_j \in (0, 1]$$

and

$$C_j = 2 \int_{x_j}^{x_j+1} A(z) dz - A(x_j) - A(x_j + 1),$$

then

$$D_j - D_{j+1} = C_j - \frac{1}{2} + e_j + e_{j+1}. \quad (55)$$

On a fully inner cell,

$$C_j = - \int_0^1 u(1-u) A''(x_j + u) du \geq 0.$$

Individual increments in (55) can nevertheless be negative, and a negative increment need not be paid by the next floor-drop cell. Thus the viable block formulation must retain the terminal convex reserve. If J is the first index with $x_J \geq \mu$, the exact target is

$$\boxed{D_0 = \sum_{j=0}^{J-1} (D_j - D_{j+1}) + D_J \geq 0,} \quad (56)$$

where $D_J \geq 0$ is the translated convex ball-tail defect. A proof of (56) needs a quantitative, not merely nonnegative, lower bound for D_J or an equivalent matched shelf reserve.

The many-cell floor dynamics can in fact be compressed to one initial shelf and one terminal one-interface tail. This is the form used below.

Proposition 7.1 (Wall-safe first-shelf reduction). *Suppose $r < \mu$ and put*

$$n = \lfloor \mu - r \rfloor, \quad q = r + n, \quad d_\rho = \frac{2 \arcsin \rho}{\pi}.$$

For $0 \leq j \leq n$, let

$$F_j = \left\lfloor A(r + j) + \frac{1}{4} \right\rfloor.$$

If this sequence leaves its first shelf, let p be the last index on that shelf; otherwise put $p = n$. Define

$$R_p = 2 \int_r^{r+p} A(z) dz - 2pF_0. \quad (57)$$

Then, at ordinary and strict floor walls alike,

$$\boxed{D_A(r) \geq D_A(q) + R_p + \frac{d_\rho}{2}(n - p).} \quad (58)$$

Moreover, if

$$\varepsilon_j = A(r + j) + \frac{1}{4} - F_j \in [0, 1), \quad \sigma = -A',$$

then

$$R_p = C_p + p \left(\varepsilon_0 + \varepsilon_p - \frac{1}{2} \right), \quad (59)$$

$$C_p := - \int_0^p u(p - u) A''(r + u) du \geq 0, \quad (60)$$

and

$$R_p \geq -\frac{p}{2} + 2 \int_0^p u \sigma(r+u) du. \quad (61)$$

In particular, with

$$\kappa = \frac{1-\rho}{\pi \rho K}, \quad a_\rho = \frac{2\pi \rho}{1-\rho},$$

one has

$$p < \sqrt{r^2 + a_\rho K} - r. \quad (62)$$

Proof. Put $h(t) = A(r+t) + 1/4$. At half-tail scale, the strict trapezoidal sum

$$\mathbf{T}_<(h; a, b) := \frac{1}{2}[h(a)]_< + \sum_{j=a+1}^{b-1} [h(j)]_< + \frac{1}{2}[h(b)]_<$$

on $[0, p]$ is at most pF_0 . If $p < n$, the ordinary floor of h drops immediately after the left endpoint of $[p, n]$. The quantitative concave trapezoidal-floor theorem [3, Theorem 1.4] gives

$$\mathbf{T}_<(h; p, n) \leq \int_p^n A(r+t) dt - \frac{d_\rho}{4}(n-p).$$

If $p = n$, this block is absent. Additivity at p and n leaves exactly one half of the strict tail beginning at q , and hence

$$\frac{T_A(r)}{2} \leq pF_0 + \int_{r+p}^q A(z) dz - \frac{d_\rho}{4}(n-p) + \frac{T_A(q)}{2}.$$

Subtracting this from $\int_r^K A$ proves (58). At an ordinary wall the strict bracket is smaller than the ordinary floor used in the majorant, so no limiting argument is needed.

Since $F_0 = F_p$, the trapezoidal identity gives (59)–(60). Also $F_0 \leq A(r+p) + 1/4$, and therefore

$$R_p \geq -\frac{p}{2} + 2 \int_0^p (A(r+t) - A(r+p)) dt,$$

which is (61) after reversing the order of integration. Finally, on the inner branch

$$\sigma'(z) = \frac{1}{\pi \sqrt{\mu^2 - z^2}} - \frac{1}{\pi \sqrt{K^2 - z^2}} \geq \kappa, \quad \sigma(0) = 0.$$

Equality of the two endpoint floors implies $A(r) - A(r+p) < 1$. Consequently

$$1 > \int_r^{r+p} \sigma(z) dz \geq \frac{\kappa}{2}((r+p)^2 - r^2),$$

which is exactly (62). $\square$

The proposition identifies the sharp scalar which remains. The point q is a one-interface start, so Theorem 6.7 gives $D_A(q) \geq 0$ and, more precisely,

$$D_A(q) = D_K(q) + \lambda_q - 2 \int_q^\mu G_\mu(z) dz, \quad \lambda_q = [G_K(q) + \tfrac{1}{4}]_< - [A(q) + \tfrac{1}{4}]_<. \quad (63)$$

Thus every tail satisfying

$$d_\rho(n-p) \geq p \quad (64)$$

is already proved by (58) and $R_p \geq -p/2$. Any residual tail must satisfy the explicit interface-band restriction

$$n < \frac{1 + d_\rho}{d_\rho} \left(\sqrt{r^2 + a_\rho K} - r \right), \quad (65)$$

and the sole remaining inequality is

$$\boxed{D_A(q) + R_p + \frac{d_\rho}{2}(n - p) \geq 0.} \quad (66)$$

Unlike a local bad-cell pairing, this scalar retains the terminal convex reserve forced by the counterexamples above.

Lemma 7.2 (Scale-free shelf curvature). *In the notation of Proposition 7.1, suppose $p \geq 1$ and put*

$$\Delta = A(r) - A(r + p).$$

Then

$$C_p \geq \frac{p^2}{3(2r + p)} \Delta. \quad (67)$$

Consequently, since $\Delta = \varepsilon_0 - \varepsilon_p$ on the first ordinary-floor shelf,

$$R_p \geq \frac{p^2}{3(2r + p)}(\varepsilon_0 - \varepsilon_p) + p \left(\varepsilon_0 + \varepsilon_p - \frac{1}{2} \right). \quad (68)$$

Moreover,

$$A(0) - A(r) \leq \frac{r\Delta}{2p}. \quad (69)$$

Proof. On the inner branch $\sigma = -A'$ is nonnegative, increasing, and convex, with $\sigma(0) = 0$. Indeed,

$$\sigma'(z) = \frac{1}{\pi\sqrt{\mu^2 - z^2}} - \frac{1}{\pi\sqrt{K^2 - z^2}},$$

and differentiating once more preserves the sign. Use the positive hinge representation

$$\sigma(z) = az + \int_{[0, \infty)} (z - s)_+ d\nu(s), \quad a \geq 0, \quad \nu \geq 0.$$

Both sides of (67) are linear in σ . For the generator $\sigma(z) = z$ their ratio is

$$\frac{p^3/6}{p(r + p/2)} = \frac{p^2}{3(2r + p)}.$$

For a hinge with $s \leq r$, the same ratio is

$$\frac{p^3/6}{p(r - s + p/2)} \geq \frac{p^2}{3(2r + p)}.$$

If $r < s < r + p$ and $L = r + p - s$, the ratio is $p - 2L/3 \geq p/3$, which is again at least the required constant. Hinges to the right contribute zero. This proves (67), and (68) follows from (59).

Finally, convexity and $\sigma(0) = 0$ imply that $\sigma(z)/z$ is nondecreasing. Hence

$$\int_0^r \sigma(z) dz \leq \frac{r}{2} \sigma(r), \quad \Delta \geq p\sigma(r),$$

which proves (69). □

Lemma 7.3 (Level-by-level convex reserve). *Let g be nonnegative, strictly decreasing, convex, and C^1 on $[0, L)$, with $g(L) = 0$, and extend it by zero. Set*

$$D_g = 2 \int_0^L g(s) \, ds - \left([g(0) + \tfrac{1}{4}]_< + 2 \sum_{j \geq 1} [g(j) + \tfrac{1}{4}]_< \right), \quad B = [g(0) + \tfrac{1}{4}]_<.$$

For $1 \leq k \leq B$, let y_k be the unique point satisfying $g(y_k) = k - 1/4$, and put $c_k = -g'(y_k)$. Then

$$D_g \geq \sum_{k=1}^B \left(\frac{1}{2c_k} - 1 \right). \quad (70)$$

Proof. Let $y(t)$ be the decreasing inverse of g , extended by zero for $t \geq g(0)$. Convexity of g makes y convex; the zero extension remains convex because its right derivative is zero. Layer cake gives

$$2 \int_0^L g = 2 \int_0^{g(0)} y(t) \, dt.$$

The level $k - 1/4$ contributes exactly $2\lceil y_k \rceil - 1 \leq 2y_k + 1$ to the strict tail, including when y_k is integral. The tangent to y at $k - 1/4$ gives

$$\int_{k-1}^k y(t) \, dt \geq y_k - \frac{1}{4} y'(k - \tfrac{1}{4}) = y_k + \frac{1}{4c_k}.$$

Summation over the active levels proves (70). At a top action wall the equality level is absent from B , exactly as required by the strict convention. $\square$

Corollary 7.4 (Exact-angle terminal reserve). *At the one-interface point q , put*

$$B = [G_K(q) + \tfrac{1}{4}]_<, \quad Q = [A(q) + \tfrac{1}{4}]_<.$$

For $1 \leq k \leq B$, let $\theta_k \in (0, \pi/2)$ be the unique solution of

$$\frac{K}{\pi} (\sin \theta_k - \theta_k \cos \theta_k) = k - \frac{1}{4}.$$

Then

$$D_A(q) \geq \max \left\{ 0, \frac{\pi}{2} \sum_{k=1}^B \frac{1}{\theta_k} - Q - 2 \int_q^\mu G_\mu(z) \, dz \right\}. \quad (71)$$

Proof. Apply Lemma 7.3 to $g(s) = G_K(q + s)$. At the level $k - 1/4$ the ball slope is $c_k = \theta_k/\pi$, so

$$D_K(q) \geq \sum_{k=1}^B \left(\frac{\pi}{2\theta_k} - 1 \right).$$

Insert this in (63); the $-B$ from the sum cancels the $+B$ in $\lambda_q = B - Q$. The other entry in the maximum is the completed one-interface theorem. $\square$

Proposition 7.5 (Exact no-drop identities). *Suppose $p = n \geq 1$, and write $F_0 = \dots = F_n = f$. Put*

$$\varepsilon_q = A(q) + \frac{1}{4} - f \in [0, 1), \quad \chi_q = \mathbf{1}_{\{A(q) + 1/4 = f\}}.$$

Then

$$D_A(r) = D_A(q) + R_n + \chi_q, \quad (72)$$

$$R_n = -\frac{n}{2} + 2n\varepsilon_q + 2 \int_0^n u \sigma(r+u) du. \quad (73)$$

Thus the no-drop branch is automatic when $f = 0$, or $\varepsilon_q \geq 1/4$, or

$$2 \int_0^n u \sigma(r+u) du \geq n \left(\frac{1}{2} - 2\varepsilon_q \right).$$

Away from the endpoint wall, the exact shifted-tail condition is the quantitative payment

$$D_A(q) \geq \frac{n}{2} - 2n\varepsilon_q - 2 \int_0^n u \sigma(r+u) du > 0. \quad (74)$$

At the endpoint wall the right side is reduced by the favorable unit $\chi_q = 1$ in (72); proving the wall-safe scalar (66) directly asks for the stronger displayed payment.

Proof. Strict decrease shows that a lower ordinary-floor wall cannot occur at an inner sample before q without ending the shelf. Hence the strict brackets at $r, \dots, q$ equal f , except that the bracket at q is $f - 1$ when $\chi_q = 1$. Therefore

$$T_A(r) - T_A(q) = 2nf - \chi_q,$$

which proves (72). Since $f = A(q) + 1/4 - \varepsilon_q$,

$$\begin{aligned} R_n &= 2 \int_0^n (A(r+t) - A(q)) dt + 2n \left(\varepsilon_q - \frac{1}{4} \right) \\ &= 2 \int_0^n \int_t^n \sigma(r+u) du dt + 2n \left(\varepsilon_q - \frac{1}{4} \right), \end{aligned}$$

and reversing the order proves (73). The remaining statements follow from $D_A(q) \geq 0$. $\square$

8 A dimension-dependent no-mode region

The radial form (6), the one-dimensional Dirichlet Poincaré inequality, and $r^{-2} \geq 1$ on $(\rho, 1)$ give

$$\lambda_{\ell,n} \geq \frac{n^2 \pi^2}{(1-\rho)^2} + \nu_{\ell,d}^2 - \frac{1}{4}.$$

Since $\nu_{0,d} = (d-2)/2$, one obtains the strengthened wall

$$N_D(A_{\rho,1}^{(d)}, K^2) = 0 \quad \text{if} \quad K^2 \leq \frac{\pi^2}{(1-\rho)^2} + \frac{(d-2)^2 - 1}{4}. \quad (75)$$

The equality face belongs to the no-mode region because the counting function is strict.

In particular, for the general-dimensional Pólya theorem it is enough to prove the shifted-tail estimate in the universal active region

$$K > \frac{\pi}{1-\rho}. \quad (76)$$

Indeed, the complementary region has no eigenvalues already in dimension three, and the additional centrifugal term only enlarges it for $d > 3$. This observation does not prove the stronger standalone Conjecture 1.1 below the wall, but it removes that region from the spectral critical path.

9 Remaining proof obligations

The first interface-cell milestone is complete, and Proposition 6.8 proves that for each fixed ρ only a bounded frequency interval remains; in the thin sector Proposition 6.10 improves the sufficient threshold to $K \geq 125/[8(1 - \rho)^2]$. Proposition 7.1 further replaces the many-cell dynamics by the single scalar (66). For the spectral theorem this scalar is needed only in (76). Lemma 7.2 sharpens its head term, while Corollary 7.4 replaces qualitative terminal nonnegativity by an explicit positive sum. Proposition 7.5 also identifies the exact endpoint phase when $p = n$. The primary route now has the following precise steps.

1. Prove (74) in the residual no-drop branch, with the favorable wall unit from (72) retained.
2. For $p < n$, combine (68) and (71) to prove (66). In the first-floor branch, the current wall certificate closes ten explicit residual chambers, but an analytic exhaustion of all other chambers is still required.
3. Insert the resulting all-frequency shifted-tail theorem into Theorem 5.3. No dimension-dependent angular or optical theorem is then needed.

If the first step fails only for an explicitly controlled family of tails, the exact identity (24) is the appropriate aggregate fallback: the weighted negative part must be compared directly with $\mathcal{B}_d(A)$.

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